LEARNER GUIDE

Loci — the shape of a rule

Constructions, Loci and Bearings · Lesson 5. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A goat on a rope can reach a shape — everywhere no further than the rope from its post. That shape is fixed by one rule, and a set of all the points obeying a rule is called a **locus** (plural **loci**). You have already met two: the perpendicular bisector (equidistant from two points) and the angle bisector (equidistant from two lines). This lesson names the idea and builds the everyday loci — a fixed distance from a point (a circle) and a fixed distance from a line (a rounded 'racetrack') — and combines two rules to find a region.

What you are learning to notice

- A **locus** is the set of all points that obey a rule.
- A fixed distance from a single **point** is a **circle**; 'within that distance' fills the disc.
- A fixed distance from a **segment** is two parallel sides with **semicircular ends**.
- Two rules together give a **region** — the overlap of the two loci.

Moving through it, one step at a time

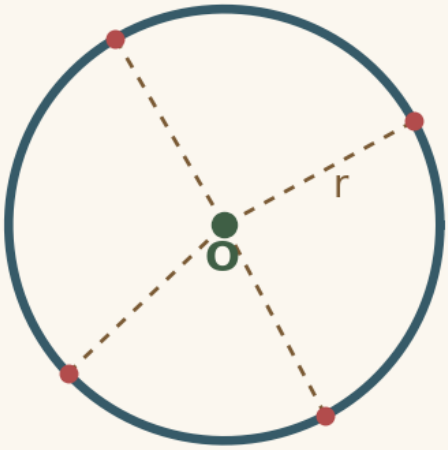
Read the rule and picture its locus. 'A fixed distance from a point' → set the compass to that distance and draw a circle. 'A fixed distance from a line' → draw two lines parallel to it, that distance either side, and join them with semicircular ends at the endpoints. For a **region** problem, draw the boundary of each rule, then shade the part that obeys them all at once — usually where the loci overlap. Leave your construction marks showing.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — The locus a fixed distance from a point

Draw the locus of all points exactly 3 cm from a point O.

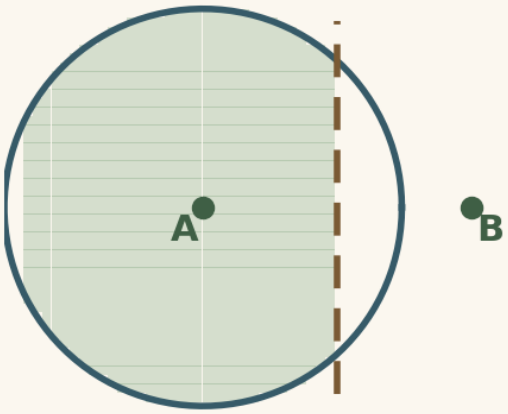


1. Mark the point O.	point O
2. Every point of the locus is 3 cm from O.	distance 3 cm
3. Set the compass to 3 cm and draw the circle.	radius 3 cm

The locus is a circle of radius 3 cm, centred O — every point on it is 3 cm from O.

Example 2 — A region from two rules

Shade the points that are within 3 cm of A AND closer to A than to B.



1. 'Within 3 cm of A' — draw a circle of radius 3 cm centred A; inside it obeys the rule.	circle at A
2. 'Closer to A than B' — construct the perpendicular bisector of AB; A's side obeys the rule.	perp. bisector
3. Shade where both are true — inside the circle and on A's side.	overlap

The region is the part of the circle on A's side of the perpendicular bisector — where the two loci overlap.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the perpendicular bisector (Lesson 3) and the angle bisector (Lesson 4)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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